

COMP CO835

Object-Oriented Systems

Week 7

Defining the System Architecture

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Chapter 7: Outline

- Anatomy of Modern Information Systems
- Architectural Concepts
- Interoperability
- Architectural Diagrams
- Three Layer Architecture
- Describing the Environment
- Designing Application Components

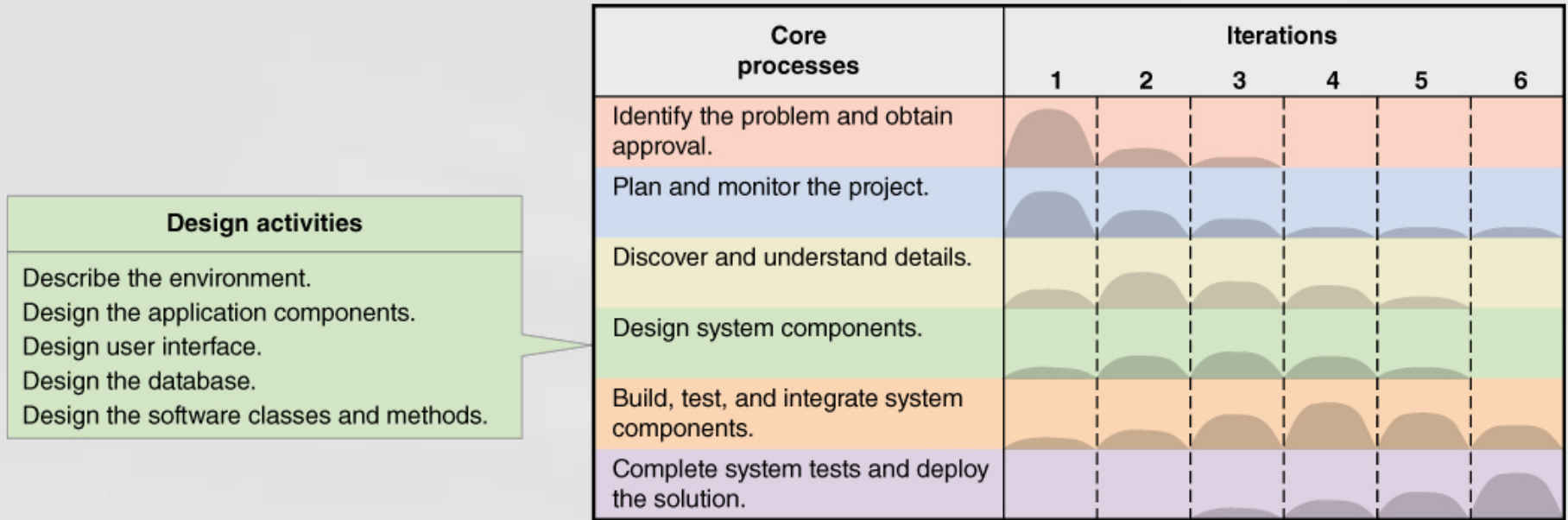
Learning Objectives

- Explain architectural concepts that influence system design, including ever-present computing and software, components, protocols, interoperability, and distributed architectures
- Describe and draw location, network, and deployment diagrams
- Describe a system's environment by drawing architectural diagrams and answering key questions
- Design larger application components based on use cases and other analysis models

Overview

- An important part of new system development is choosing appropriate technologies
- Explain and provide a summary of technology and architectural concepts
- describe the details for the activity – *Describe the Environment*
- Describe the details for the activity – *Design the application components*

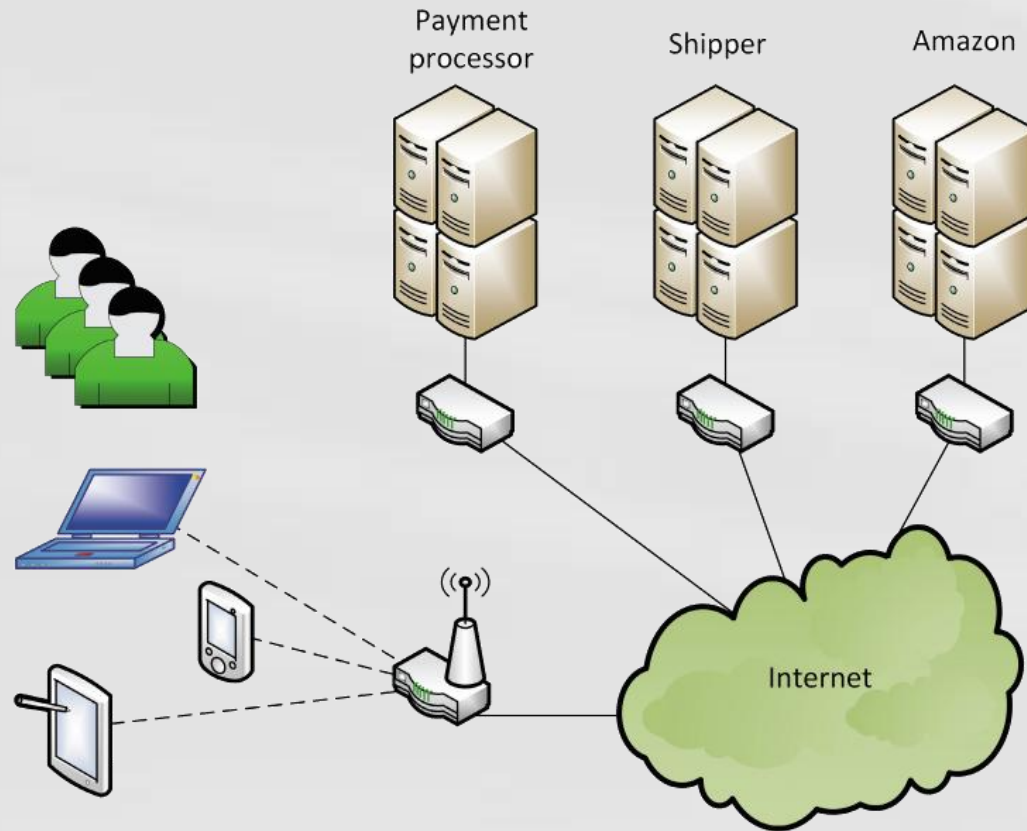
Activities of “Design System Components”



Anatomy of a Modern System – Computing Devices

- Server – manages shared resources and enables users and other computers access to these resources
- Personal computing devices or clients
 - Desktops, laptops, tablets, smartphones...

Simplified architecture for application (Amazon.com)



Server Farm –

- Very large databases and very high use

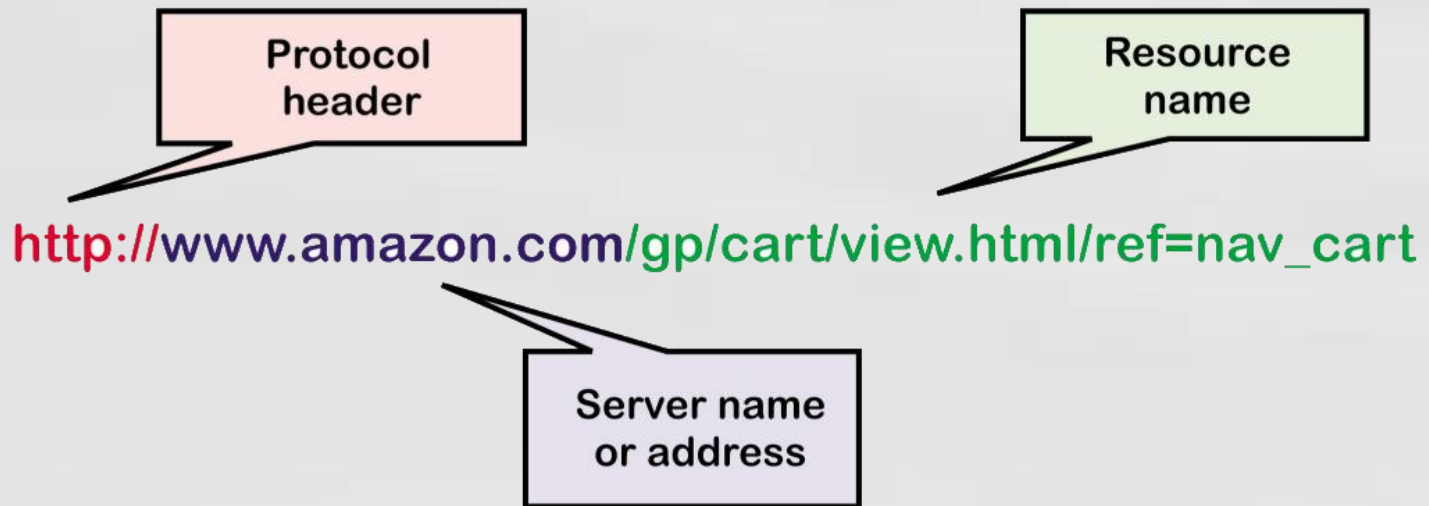


Anatomy - Networks (1 of 2)

- Computer network – hardware, software, transmission media
- Internet backbone –
 - High-capacity with high-bandwidth trunk lines and large high-speed computers
 - Owned by governments and telecom companies
- Local area network (LAN) –
 - Small network for a single site
- World Wide Web (WWW)
 - All the interconnected resources accessed through the Internet

Anatomy - Networks (2 of 2)

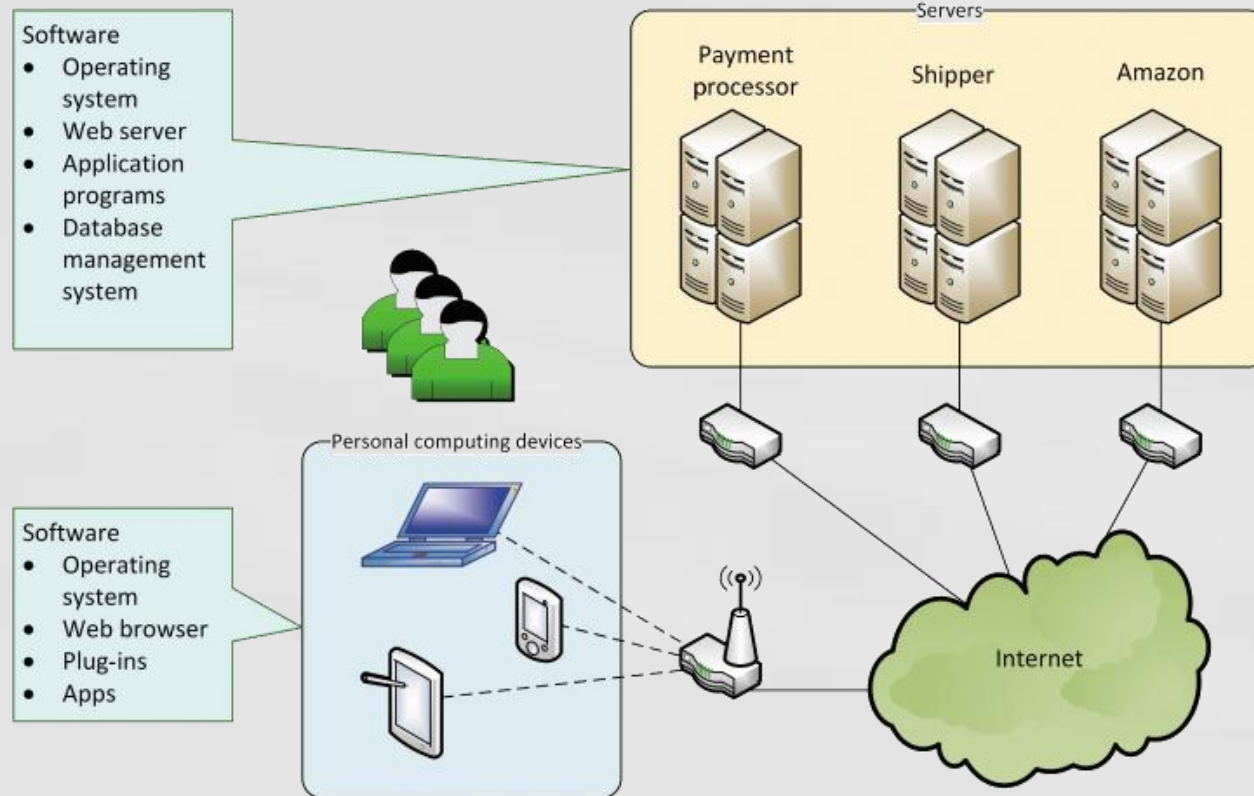
- Uniform Resource Locator (URL)
 - The identifier for the Web to locate a particular resource
- Hyperlink –
 - The URL of a resource embedded within another resource



Anatomy – Software (1 of 3)

- Application software – programs that perform work for users
 - Either a custom app or a Web-based application
- App –
 - A custom program usually for a laptop or smartphone
- System Software –
 - Behind the scene software, works as glue to hold everything together

Anatomy – Software (2 of 3)



Anatomy – Software (3 of 3)

• Web-Based Applications

- Uses a web browser
- Accessed through a URL
- Resides on a Web server
- Uses standard IP protocols

• Embedded Software

- Software apps or functions embedded within another app, such as within a browser or O/S
- Toolbars, Plug-ins, Widgets

Anatomy – Protocols

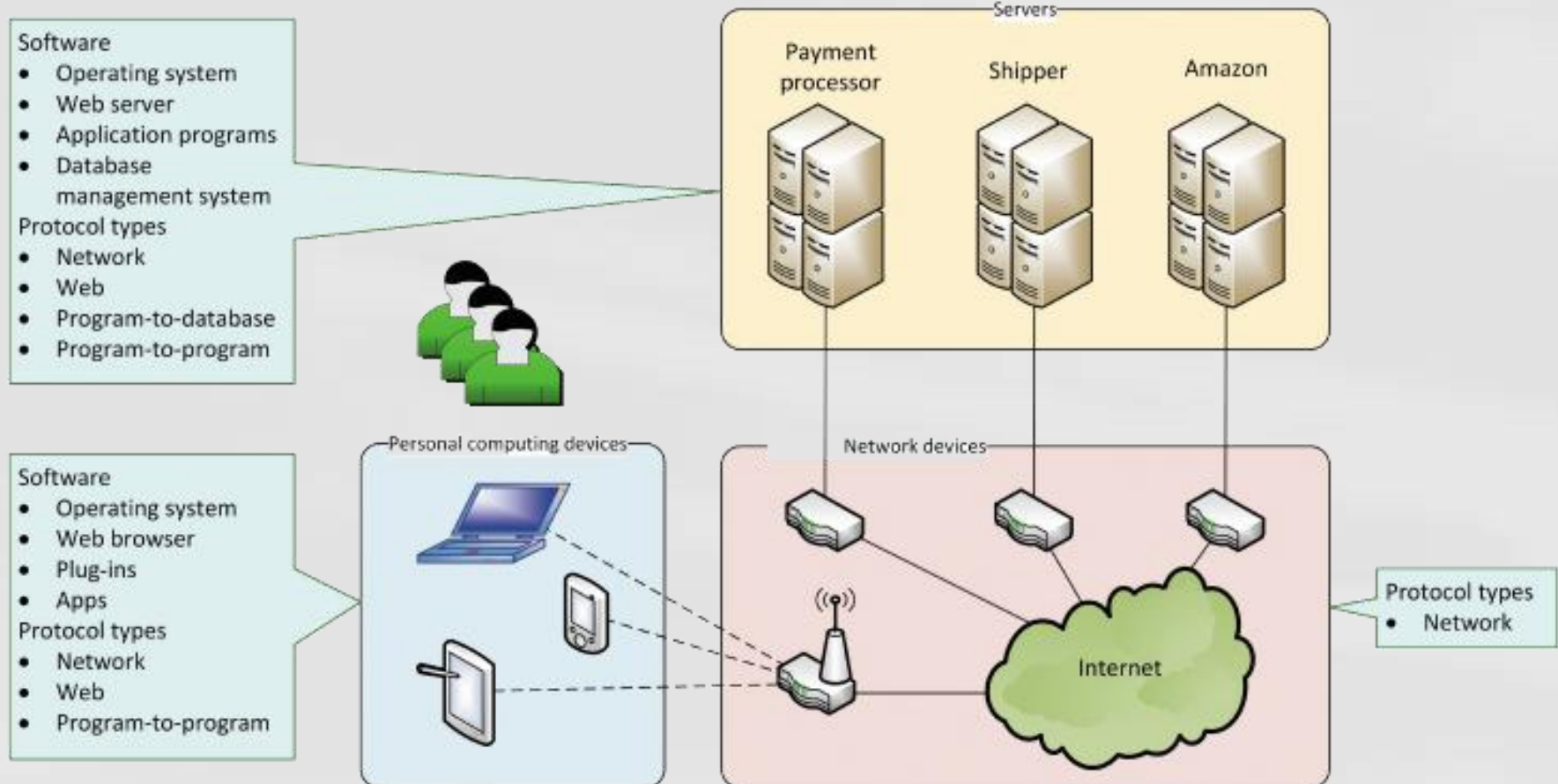
- Protocol

- A set of languages and rules to ensure communication and data exchange between hardware and software

- Network protocols

- Virtual Private Network (VPN)
 - Creates a private network but on the Internet by using secure technologies and encryption

Anatomy – Software and Protocols



Anatomy – Web Protocols

- HTML (hypertext markup language)
 - Protocol for the structure and content of a Web page
- XML (extensible markup language)
 - An extensions of HTML that enables defining semantics of tags
- HTTP (hypertext transfer protocol)
 - Defines format and content for transfer of Web documents
- HTTPS (hypertext transfer protocol secure)
 - Encrypted and secure http transfers

Architectural Concepts

- Technology architecture
 - Computers, network computers and hardware, and system software
- Application architecture
 - The software programs and their configuration

Software as a Service (SaaS)

• SaaS

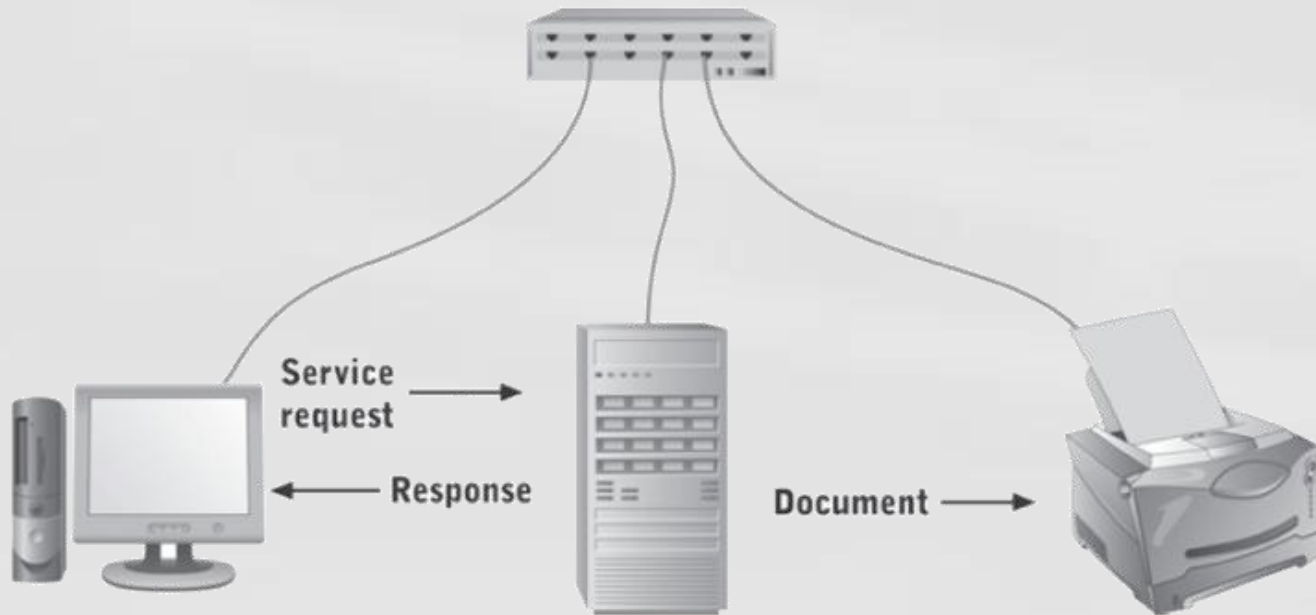
- No software is installed on the user's device
- Application services is accessed remotely
- User data is isolated and stored on common servers

Web Services

- Web service
 - Software function that is executed with Web standards
 - Access via a URL
 - Inputs sent via the URL
 - Executes remotely
 - Data returned within a Web page

Distributed Architectures

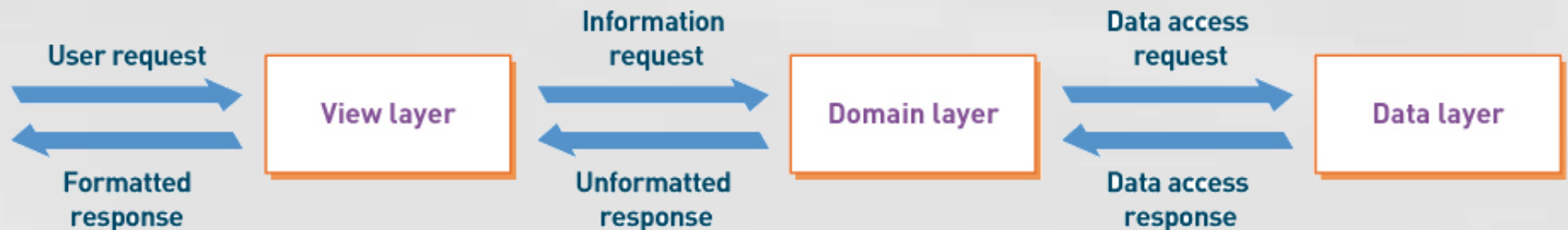
- Client/Server architecture
 - Software design with part of the application on a server and part on the client



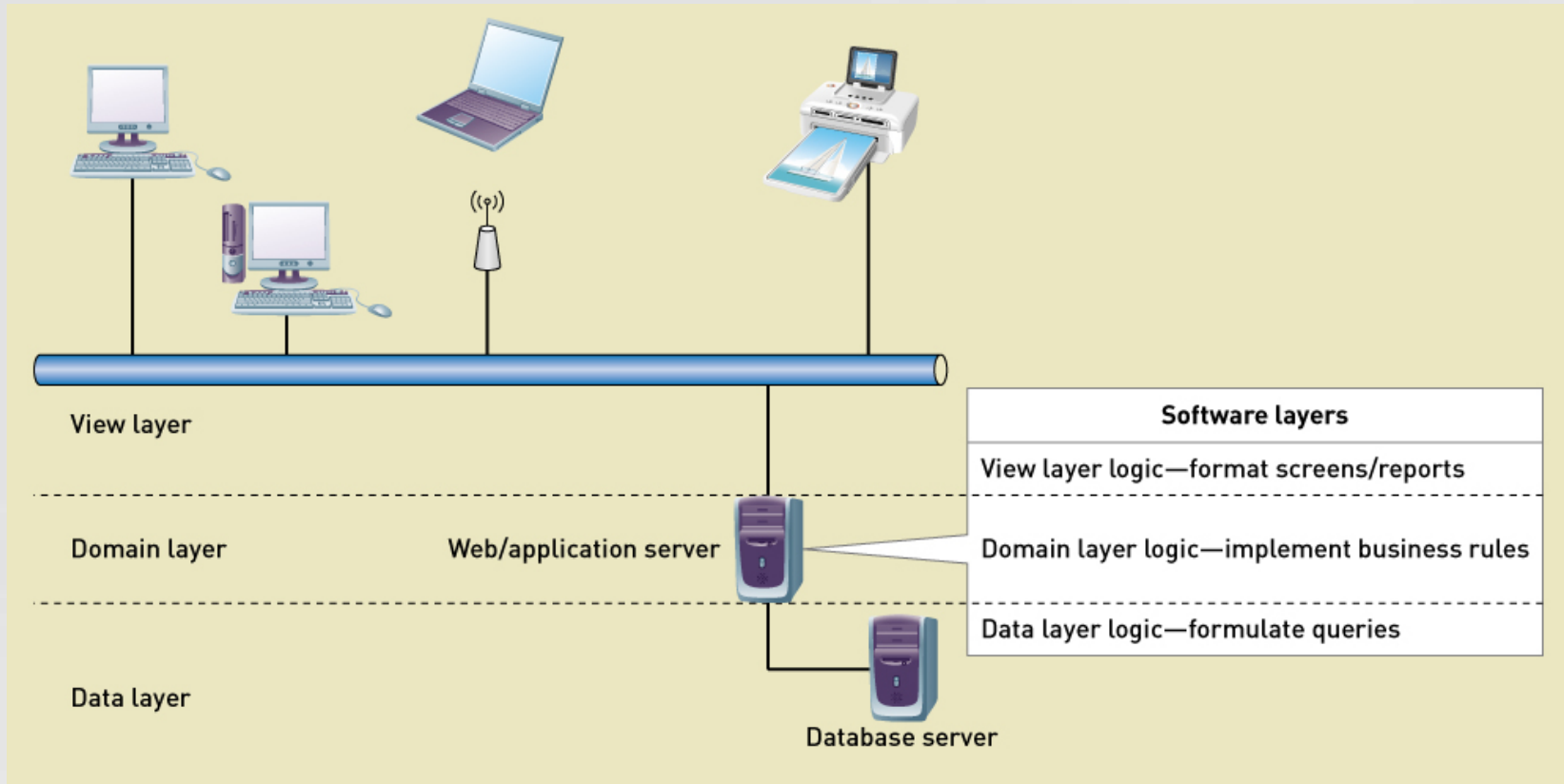
Distributed Architecture

• Three-Layer architecture

- Client/server architecture with application divided into view layer, logic layer, and data layer
- View layer – the user interface
- logic layer – program logic to implement the functions
- data layer – the functions to access the data



Three Layer Architecture



The three-layer architecture

- The three-layer architecture promotes separation of concerns, modularity, and maintainability, making it easier to develop, test, and scale software systems.
- It also allows for easier collaboration among teams working on different parts of the system, as each layer can be developed and maintained independently

Presentation Layer (View Layer)

- This is the topmost layer and is responsible for interacting with users. It focuses on the user interface (UI) and user experience (UX).
- Components in this layer include web browsers, mobile apps, desktop applications, or any other interface through which users interact with the system.
- The primary concern of this layer is to display information to users and capture user input. It doesn't contain any business logic but delegates user requests to the business logic layer for processing.
- The presentation layer is often designed to be independent of the underlying business logic and data access layers, allowing for easier updates and changes to the UI without affecting other parts of the system.

Business Logic Layer (Application/Domain Layer):

- The business logic layer sits between the presentation and data access layers.
- It contains the core functionality and rules of the application. This is where business logic, algorithms, and processing logic reside.
- The primary responsibilities of this layer include processing user inputs from the presentation layer, applying business rules, performing computations, and generating appropriate responses.
- It encapsulates the logic necessary to implement use cases and business processes without being concerned with how data is stored or retrieved.

Data Access Layer (Data Layer)

- The data access layer is responsible for interacting with the underlying data storage systems, such as databases, file systems, or external APIs.
- It abstracts the details of data storage and retrieval from the business logic layer, providing a consistent interface for accessing and manipulating data.
- This layer typically includes database management systems (DBMS) and other data access technologies.
- By separating data access concerns into a distinct layer, the system becomes more maintainable, as changes to data storage mechanisms can be made without affecting the business logic layer or the presentation layer.

Interoperability

- Interoperability

- The ability of an application to interact with other software

- Important characteristic in current development projects

- Understand the environment
 - Reuse software existing components (purchased or in-house)
 - Build components considering interoperability
 - Combine all components into a solution system

Diagrams for System Architectures

- Location Diagrams

- Identify geographical placement of hardware, software, and users

- A Location Diagram, in the context of system design or architecture, is a visual representation that illustrates the physical or logical distribution of components within a system or an application.

- It provides an overview of the various components and their interconnections across different locations or environments.

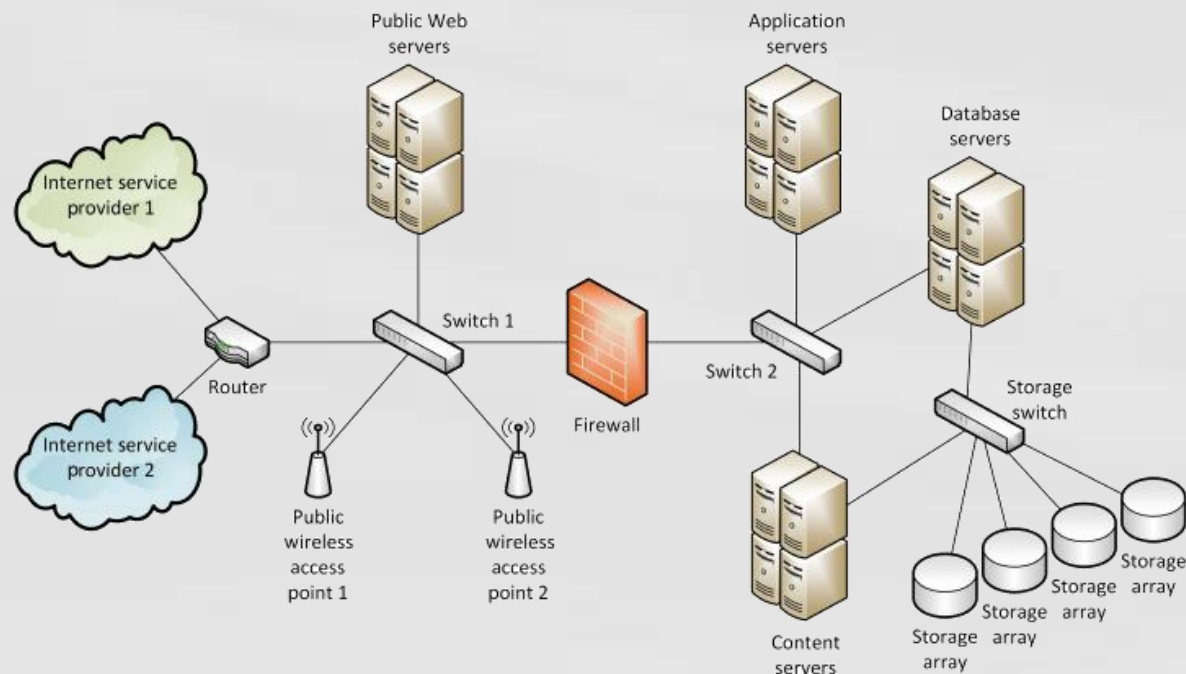
Location Diagram



Diagrams for System Architecture (1 of 2)

Network Diagrams

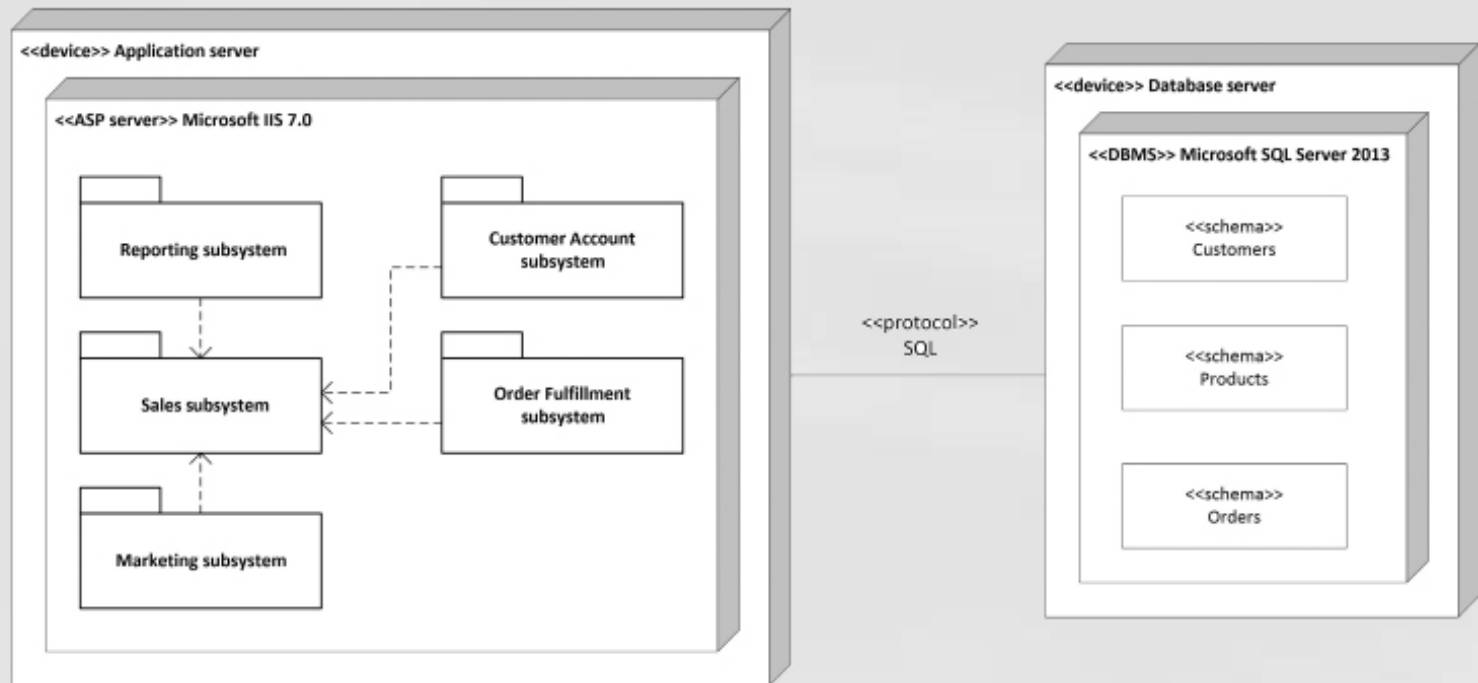
- How the application software is deployed across the hardware and system software



Diagrams for System Architecture (2 of 2)

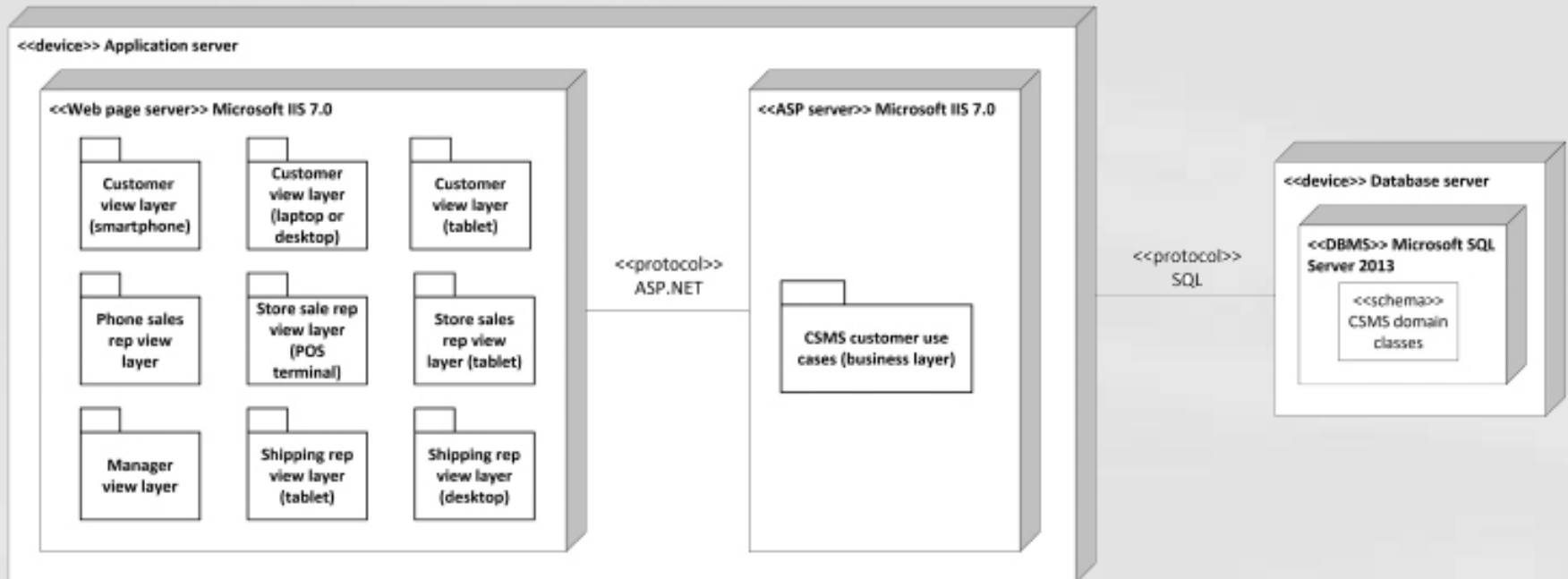
Deployment Diagrams

- How the components of a network are interconnected



RMO CSMS Deployment Diagram

- Three-layer design with user components grouped by user functions



System Design Overview

Describe the Environment

- Describing the environment in system design refers to providing a comprehensive understanding of the context in which the system will operate.
- This includes various factors such as the physical infrastructure, technological landscape, organizational policies, user demographics, and any other external influences that may impact the system's design, development, deployment, and operation.

Describing the Environment (1 of 2)

- Key Questions to help describe accurately
 - What are the key features of existing or new environment
 - O/S, system software, networks, tools
 - What are the external systems or DBMSs
 - What kind of interaction
 - What is the data
 - What are the protocols
 - What kind of security

Describing the Environment (2 of 2)

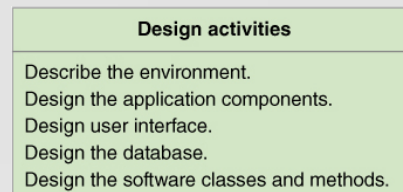
- Key Questions to help describe accurately
 - What devices will be required
 - Protocols for devices
 - Security
 - What APIs
 - What user-interface technology will be used
 - Where and who are users, and what skills
 - What hardware and devices
 - What client O/S will be used
 - Security requirements
 - What APIs are needed

Designing Application Components

● Application Component Boundaries

- Which components perform which functions
- How to group functions to build components
 - Actors – what functions do particular actors use
 - Shared data – what functions use the same data
 - Events – what functions occur in everyday business events

● The three-layer architecture



Core processes	Iterations					
	1	2	3	4	5	6
Identify the problem and obtain approval.						
Plan and monitor the project.						
Discover and understand details.						
Design system components.						
Build, test, and integrate system components.						
Complete system tests and deploy the solution.						

Summary (1 of 3)

- Anatomy of a Modern Information System
 - Consist of computing devices, networks, software, and protocols
 - Deployed as stand-alone software, network based, Web based
- Architectural Concepts
 - SaaS – software as a service
 - Web services
 - Distributed architectures
 - Client/server and three-layer architecture

Summary (2 of 3)

- Interoperability
 - Getting all the components to work together
- Architectural diagrams
 - Location diagrams
 - Network diagrams
 - Deployment diagrams

Summary (3 of 3)

• Describing the Environment

- External systems
- Technology architecture
- Key questions requiring answers

• Designing Application Components

- Application component boundaries
- Grouping functions into components
- System of record – who owns the data

Questions?